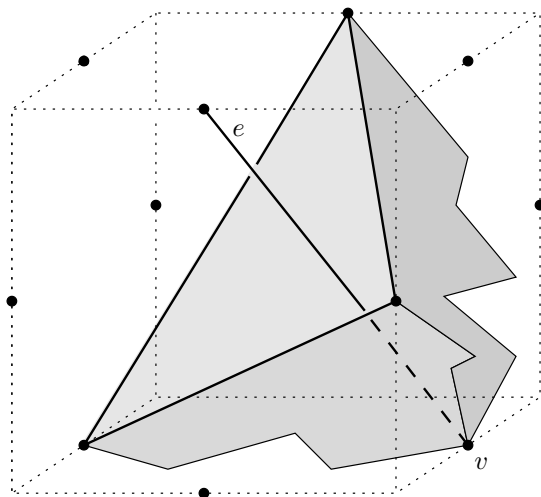


Now assume that the resulting simplicial complex $\mathcal{C}$ is shellable. We start with one tetrahedron F_1 , and then add new ones, $F_2, F_3, \dots$. At every step, except possibly the first, not more than one new edge from the set E^{12} can be added, since the edges in one of the triangles cannot be in a common simplex, and if two skew edges were new, then this would contradict a characteristic property of shellings of simplicial complexes: there has to be a unique minimal new face at each step, which is nonempty after the first step (see Exercise 8.2).

Now let F_{j+1} be the tetrahedron by which the *last* edge e from E^{12} is added. At this point of the shelling the complex $\mathcal{C}_j := \mathcal{C}(F_1, \dots, F_j)$ already contains the three edges e_1, e_2, e_3 that surround e . In fact, the circle formed in $\mathcal{C}_j$ by $C := e_1 \cup e_2 \cup e_3$ can be contracted within the complex $\mathcal{C}_j$: this is a property of shellable complexes of dimension at least 2, which is easy to verify by induction (they are “simply connected”).



Since the circle C surrounds the edge e , we have to “pass over” a vertex $v \subseteq e$ when we contract C in $\mathcal{C}_j$. Also, the link of this vertex v in the complex $\mathcal{C}_j$, $\text{link}(v, \mathcal{C}_j)$, is a 2-dimensional shellable complex, by Lemma 8.7. However, we can contract our circle C within $|\mathcal{C}_j|$ until it lies in $\text{link}(v, \mathcal{C}_j)$, but it cannot be contracted within the link, because then it would not pass over v . This shows that the link is not shellable: contradiction. $\square$

8.2 Shelling Polytopes

The complex of a polytope $\mathcal{C}(P)$ is shellable if and only if the boundary complex $\mathcal{C}(\partial P)$ is shellable. Thus, “shelling a polytope” means finding a shelling order for the facets of P . How do we do this? Is there an obvious

way? If we polarize, then the problem is to find a good ordering on the vertices of the polar polytope P^Δ . For this an obvious thing to do is to take a linear function in general position (see Sections 3.1 and 3.4), and to order the vertices according to that linear function. This works: it does yield a shelling (in fact, many shellings) for the boundary of P .

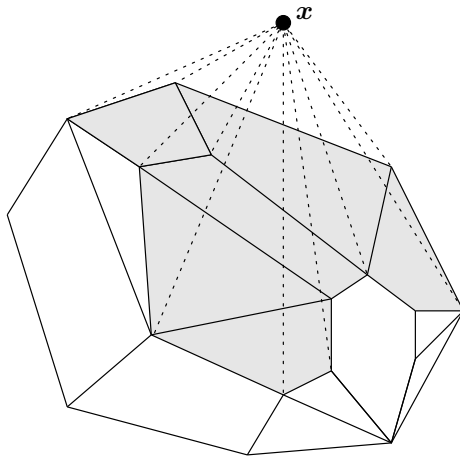
Below we describe these shellings directly on P , without polarization. In Exercise 8.10 you are asked to check that the constructions on P and P^Δ really are equivalent.

Lemma 8.10. *If $F_1, F_2, \dots, F_s$ is a shelling order for the boundary of a polytope P , then so is the reverse order $F_s, F_{s-1}, \dots, F_1$.*

Proof. Let F_j be one of the facets in the shelling, then for every facet G of F_j there is a unique other facet F_i of P such that $G = F_j \cap F_i$. This other facet F_i can be either earlier ($i < j$) or later ($i > j$) than F_j . These roles are interchanged if, while reversing the shelling of ∂P , we also reverse the shellings of the boundaries of its facets: and this we may do, by induction on the dimension. $\square$

Theorem 8.11. (Bruggesser & Mani [139]) *Polytopes are shellable.*

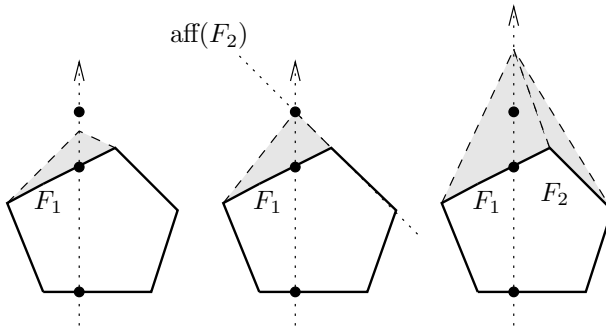
Well, this is the essence. However, what one actually needs is that there are shellings with very specific properties. These are obtained from the Bruggesser-Mani construction, which yields the much more specific theorem below (which includes later refinements by McMullen [389], Danaraj & Klee [171], and by Björner & Wachs [98]). Also, this is the technical statement that has an easy proof by induction on the dimension.



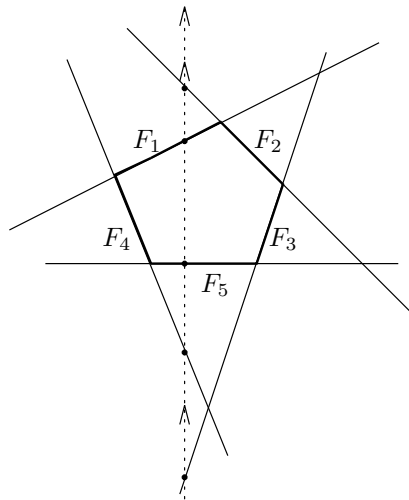
Theorem 8.12. [139] *Let $P \subseteq \mathbb{R}^d$ be a d -polytope, and let $x \in \mathbb{R}^d$ be a point outside P . If x lies in general position (that is, not in the affine hull of a facet of P), then the boundary complex $\mathcal{C}(\partial P)$ has a shelling in which the facets of P that are visible from x come first.*

Here we can use our intuition to understand what *visible* means: a facet $F \subseteq P$ is visible from $\mathbf{x}$ if for every $\mathbf{y} \in F$ the closed line segment $[\mathbf{x}, \mathbf{y}]$ intersects P only in the point $\mathbf{y}$. Equivalently, F is visible from $\mathbf{x}$ if and only if $\mathbf{x}$ and $\text{int}(P)$ are on different sides of the hyperplane $\text{aff}(F)$ spanned by F . For example, if $\mathbf{x}_G$ is beyond the face G (in the sense of Section 3.1), then the facets that contain G are exactly those that are visible from $\mathbf{x}_G$.

Proof. Given $\mathbf{x}$, we choose a line ℓ through $\mathbf{x}$ and through a point in general position in P . The properties we need are that ℓ contains $\mathbf{x}$, hits the interior of P , and the intersection points with the facet hyperplanes $\ell \cap \text{aff}(F)$ are distinct. For simplicity, assume that the line is not parallel to any of the facet hyperplanes, so we have no intersection point “at infinity.” We orient the line ℓ from P to $\mathbf{x}$.



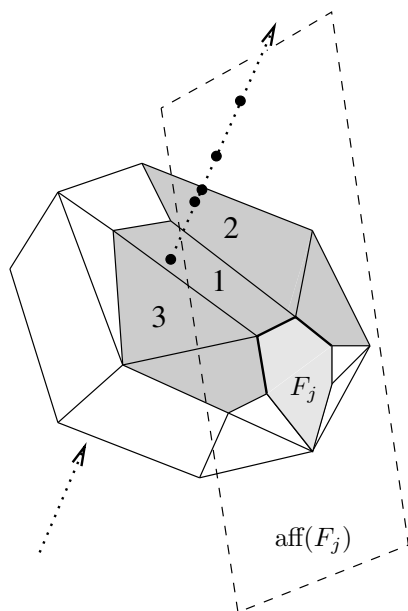
Now imagine P to be a little polyhedral planet, and have a rocket start on its surface at the point where the oriented line ℓ leaves the planet. This point lies on a unique facet F_1 , and for the first few minutes of the flight only this one facet F_1 is visible from the rocket.



After a while, a new facet will appear on the horizon: the rocket passes through a hyperplane $\text{aff}(F_2)$, and we label the corresponding facet F_2 . Continuing this, we label the facets $F_3, F_4, \dots$ in the order in which the rocket passes through their hyperplanes, that is, in the order in which the facets appear on the horizon, becoming visible from the rocket. Now we pass through infinity, and imagine that the rocket comes back to the planet from the opposite side. We continue the shelling by taking the facets in the order in which we pass through the hyperplanes $\text{aff}(F_i)$, that is, in the order in which the corresponding facets disappear on the horizon.

This “rocket flight” clearly gives us a well-defined ordering on the whole set of facets. Also, the facets that are visible from $\mathbf{x}$ form a beginning segment, since we see exactly those at the point where the rocket passes through $\mathbf{x}$.

To see that the ordering is a shelling, we consider the intersection $\partial F_j \cap (F_1 \cup \dots \cup F_{j-1})$. If F_j is added before we pass through infinity, then this intersection is exactly the set of those facets of F_j that are visible from the point $\ell \cap \text{aff}(F_j)$, at which F_j appears on the horizon. Thus, we know by induction on the dimension that this collection of facets of F_j is shellable, and can be continued to a shelling of the whole boundary ∂F_j .



After passing through infinity, the intersection is the family of nonvisible facets. This is shellable, too, because reversing the orientation of the line ℓ yields the shelling with the reversed ordering of the facets. $\square$

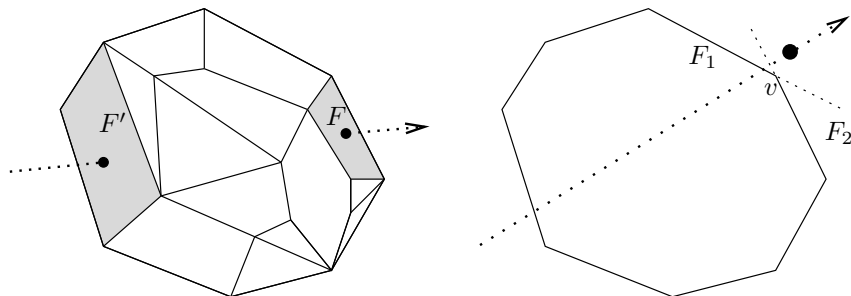
Shellings that arise by Bruggesser & Mani’s construction are also known as *line shellings*. Note that in this construction, reversing the orientation

of the line ℓ also reverses the line shelling. Thus the reverse of every line shelling is not only a shelling (by Lemma 8.10), but a line shelling as well.

The Bruggesser-Mani construction has a lot of flexibility: we can get shellings with special properties by suitable choice of the shelling line ℓ .

Corollary 8.13. *For any two facets F and F' of a polytope P , there is a shelling of ∂P in which F is the first facet and F' is the last one.*

For every vertex v of P , there is a shelling such that the facets that contain v form a beginning segment of the shelling, that is, the star of the vertex v is shelled first.



Proof. For the first claim, choose any shelling line ℓ which intersects the boundary of P in the facets F and F' . (For example, choose $\mathbf{x}$ beyond F , choose $\mathbf{x}'$ beyond F' , and let ℓ be the line determined by $\mathbf{x}$ and $\mathbf{x}'$. Perturb ℓ to general position, if necessary.)

For the second claim, let $\mathbf{x}_v$ be a point beyond the vertex v , and choose the shelling line to contain this $\mathbf{x}_v$. $\square$

Corollary 8.14. *Every Schlegel diagram is shellable.*

More generally, every regular subdivision of a polytope is shellable.

Proof. For any Schlegel diagram $\mathcal{D}(P, F)$, choose a shelling of the polytope P such that the facet F comes last. Thus the shelling of P also induces a shelling of the Schlegel diagram $\mathcal{D}(P, F)$.

Every regular subdivision $\Sigma_P(Q)$ of a d -polytope Q is, by Definition 5.3, isomorphic to the complex of faces of a $(d+1)$ -polytope P that are visible from a certain point $\mathbf{x} = -Te_{d+1}$: and thus we can apply Theorem 8.12. $\square$

This corollary applies to piles of cubes (Example 8.2(v)), for example. It suggests that every d -diagram is shellable. Is that true? (Problem 8.3*)

Knowing that polytopes are shellable, one might ask whether they are extendably shellable. Here we answer this question in the negative, and give in fact several proofs.

Theorem 8.15. *Not all 4-polytopes are extendably shellable.*

In particular, $\tilde{\mathcal{P}}_4(7, 7, 3)$ and $\tilde{\mathcal{P}}_4(7, 5, 5)$ are not extendably shellable.

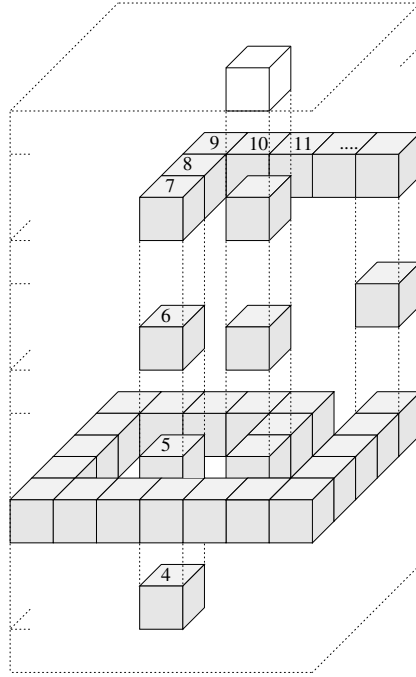
Proof. (1) Consider the 4-polytope $P_4 := \tilde{\mathcal{P}}_4(9, 9, 4)$, which contains a combinatorially equivalent copy of the pile of cubes $\mathcal{P}_3(9, 9, 4)$ on its boundary. Now we start a shelling of ∂P_4 with the facets that correspond to the nonextendable partial shelling of $\mathcal{P}_3(9, 9, 4)$ given in Lemma 8.6. This partial shelling has two connected parts in common with the boundary of the box $B := |\mathcal{P}_3(9, 9, 4)|$. Any facet of P_4 that we can add to continue the partial shelling of ∂P_4 will be connected to one of these parts of the boundary of the box, but not to both. Thus, the “missing cubes” from the box cannot be added to the partial shelling of ∂P_4 at a later time, either. This proves that $\tilde{\mathcal{P}}_4(9, 9, 4)$ is not extendably shellable. In fact, it also proves that any 4-polytope that contains an isomorphic copy of $\mathcal{P}_3(9, 9, 4)$ in its boundary complex is not extendably shellable. So, $\tilde{\mathcal{P}}_4(z_1, z_2, z_3)$ is not extendably shellable for $z_1 \geq 9$, $z_2 \geq 9$, and $z_3 \geq 4$.

(2) We delete the “bottom layer” and the “walls” of the pile $\mathcal{P}_3(9, 9, 4)$, to obtain $\mathcal{P}_3(7, 7, 3)$. Now consider the Schlegel diagram of $\tilde{\mathcal{P}}_4(7, 7, 3)$, and start shelling it by first shelling the 5 facets below and next to the pile of cubes. (This replaces the little cubes in the bottom and the walls of our previous example.) We continue the shelling by the 49 little cubes that correspond to the interior cubes in the not-extendable configuration of Lemma 8.6. If we lift this partial shelling of the Schlegel diagram to the boundary of $\tilde{\mathcal{P}}_4(7, 7, 3)$, then we can also add the facet on which the diagram was based: after that we are stuck. Thus $\tilde{\mathcal{P}}_4(7, 7, 3)$ is not extendably shellable.

(3) Start a shelling of $P'_4 := \tilde{\mathcal{P}}_4(7, 5, 5)$ at the facet F_1 that is completely disjoint from the pile of cubes. Then the remaining facets are the ones in the Schlegel diagram $\mathcal{D}(P'_4, F_1)$. Of this Schlegel diagram we next add the bottom and the top facet to the shelling, as F_2 and F_3 . Then, from the pile of cubes isomorphic to $\mathcal{P}_3(7, 5, 5)$, we take little cubes $F_4, F_5, \dots$ along a knotted curve that connects the bottom facet to the top facet of the Schlegel diagram, as suggested by the drawing on the next page. All this yields legal shelling steps, until one reaches the last (white) cube, which is in the top layer of the pile of boxes.

However, now the complex contains a knotted curve, for which all edges except for one are on the boundary of the subcomplex that we have shelled at this point. This knotted curve would be completed by the little white cube from the top layer. Now if we continue the shelling elsewhere, then there remains to be a knotted curve of this type: so the little white cube in the drawing cannot be added to the shelling in a later step, either. Thus this partial shelling of $\tilde{\mathcal{P}}_4(7, 5, 5)$ cannot be completed. $\square$

The third proof for Theorem 8.15 can also be adapted to see that neither simple polytopes, nor simplicial polytopes, are extendably shellable in general. Also, it can be generalized to see that “most” 4-polytopes are not extendably shellable. We refer to [575].



Shellings of polytopes allow us to “build up” the boundary of a polytope step by step, adding one facet at a time. Thus one can do proofs by induction on the number j of facets in the complex

$$\mathcal{C}_j := \mathcal{C}(F_1 \cup F_2 \cup \cdots \cup F_j),$$

which represents a shellable part of the boundary of a polytope.

Our first, simple but classical, application of this technique will prove the Euler-Poincaré formula.

Definition 8.16. The f -vector of a d -dimensional polyhedral complex $\mathcal{C}$ is the vector

$$\mathbf{f}(\mathcal{C}) = (f_{-1}, f_0, f_1, \dots, f_d) \in \mathbb{N}^{d+2},$$

where $f_k = f_k(\mathcal{C})$ denotes the number of k -dimensional faces in $\mathcal{C}$.

By the f -vector of a d -polytope we mean the f -vector of its boundary complex:

$$\mathbf{f}(P) := \mathbf{f}(\mathcal{C}(\partial P)) = (f_{-1}, f_0, f_1, \dots, f_{d-1}) \in \mathbb{N}^{d+1}.$$

Note that all the f -vectors we consider start with the entry $f_{-1} = 1$, corresponding to the empty face. The f -vectors of polytopes satisfy only one nontrivial linear equation: the Euler-Poincaré formula.

Corollary 8.17 (Euler-Poincaré formula).

For every d -dimensional polytope,

$$f_0 - f_1 + \cdots + (-1)^{d-1} f_{d-1} = 1 - (-1)^d.$$

Proof. In order to do induction, we need to consider the alternating sum of face numbers for general polytopal complexes. For this we define the (*reduced*) Euler characteristic $\chi(\mathcal{D})$ of a polyhedral complex of dimension at most d , by

$$\chi(\mathcal{D}) := -f_{-1} + f_0 - f_1 + \cdots + (-1)^d f_d.$$

Now if $\mathcal{D}$ and $\mathcal{D}'$ are polytopal complexes such that the union is a polytopal complex too (that is, if $F \cap F' \in \mathcal{D} \cap \mathcal{D}'$ for $F \in \mathcal{D}$, $F' \in \mathcal{D}'$), then the Euler characteristic is additive:

$$\chi(\mathcal{D}) + \chi(\mathcal{D}') = \chi(\mathcal{D} \cup \mathcal{D}') + \chi(\mathcal{D} \cap \mathcal{D}').$$

We prove by induction on d that the complex of a polytope $\mathcal{C}(P)$ always has Euler characteristic 0, and thus its boundary has Euler characteristic $(-1)^{d-1}$:

$$\chi(\mathcal{C}(P)) = 0, \quad \chi(\mathcal{C}(\partial P)) = (-1)^{d-1}.$$

This is clear for $d \leq 1$. Now if P is a d -polytope with shelling order $F_1, F_2, \dots$, then we have more precisely that

$$\chi(\mathcal{C}(F_1 \cup F_2 \cup \cdots \cup F_j)) = \begin{cases} 0 & \text{for } 1 \leq j < f_{d-1} \\ (-1)^{d-1} & \text{for } j = f_{d-1} \end{cases}$$

— which follows by induction on j and dimension, since the facets F_j that we add in are $(d-1)$ -polytopes, the Euler characteristic is additive, and the intersection $F_j \cap (\bigcup_{i < j} F_i)$ is a shellable part of, but not the whole boundary of F_j , for $j < f_{d-1}$. This last fact is immediate from Lemma 8.10, or geometrically in the special case of a line shelling. $\square$

8.3 h -Vectors and Dehn-Sommerville Equations

From now on, to the end of this lecture, all polytopes are simplicial.

For simplicial polytopes, the combinatorics of shellings can be described even more concretely. For the following let P be a simplicial d -polytope, so its boundary is a simplicial complex $\mathcal{C} := \mathcal{C}(\partial P)$ of dimension $d-1$. Let $V := \text{vert}(P)$ be the vertex set of P of size $n := f_0(P)$.

We identify the proper faces of P with their vertex sets — that is, we identify the “geometric simplicial complex $\mathcal{C}$ ” with the “abstract simplicial